

**Fourth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

ANALOG AND DIGITAL COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1.
 - (a) A carrier power 400 W is modulated on depth 75%. Determine total power in DSBFC, DSBSC and SSB. Comment on the results. 5(CO1)
 - (b) An equation of frequency modulation wave given by,
 $E_{FM} = 20 \sin (6 \times 10^2 t + 7 \sin 1250 t)$. Determine carrier frequency, modulation frequency, modulation index, power dissipated by FM wave for 100Ω resistance. 5(CO1)
 2. Illustrate the effect of White Noise in :
 - (1) Double Sideband Suppressed Carrier (DSBSC).
 - (2) Single Sideband Suppressed Carrier (SSBSC). 10(CO2)
 3.
 - (a) A TV signal with bandwidth 4.2 MHz is transmitted using PCM. The number of quantization level 512. Determine code word length, transmission bandwidth, bit rate, signal to noise ratio and signal to noise ratio for sinusoidal input. 5(CO3)
 - (b) What is disadvantage of PCM ? Explain in detail Delta modulation. 5(CO3)

4. Represent waveforms for the Transmitter output if input sequence 11100001 for following modulation techniques :
- (1) Binary Phase Shift Keying (BPSK).
 - (2) Differential Phase Shift Keying (DPSK).
 - (3) Quadrature Phase Shift Keying (QPSK).
 - (4) Minimum Shift Keying (MSK). 10(CO4)
5. A Convolution Code is describe by $X1 = [1 \ 1 \ 0]$; $X2 = [1 \ 1 \ 1]$;
Draw the convolution encoder and state diagram. If the received sequence for above encoder is 110010, decode the sequence using Viterbi decoding algorithm. 10(CO4)
6. Describe following in detail :
- (I) Inter Symbol Interference.
 - (II) Probability of error for Frequency Shift keying. 10(CO5)

